Tony Wu

Massachusetts Institute of Technology | Mathematics and Computer Science

Education

Massachusetts Institute of Technology Cambridge, MA
B.S. in Mathematics and Computer Science *Sep 2023 - May 2027 (expected)*
Cumulative GPA: 4.72/5.0

Research Interests

Theoretical computer science, statistical machine learning, and AI safety, with research experience in learning from dynamics, sigmoid-circuit lower bounds, and mechanistic estimation for neural networks.

Publications

Learning Gaussian Graphical Models from a Glauber Trajectory Without Mixing.

Eric Shen*, **Tony Wu***, Mahbod Majid, and Ankur Moitra. **Equal first authors.*

International Conference on Machine Learning (ICML), 2026. Poster. arXiv:2606.31230

Manuscripts under Review

Improved Rates of Learning Gaussian Graphical Models from Glauber Dynamics.

Tony Wu, Eric Shen, and Mahbod Majid.

Submitted to ICLR 2027.

Research Experience

Massachusetts Institute of Technology Cambridge, MA
Researcher, Circuit Complexity *Sep 2025 - Present*
Advisor: Daniel Reichmann

- Study lower bounds for sigmoid circuits in theoretical computer science.
- Design circuit constructions for Boolean functions to investigate their computational complexity.

Alignment Research Center Berkeley, CA
Visiting Researcher *Jun - Aug 2026*

- Theoretically classified when mechanistic output estimation is possible or impossible depending on how adversarial the model is, focusing on multilayer perceptrons with shifted weights.
- Developed estimation algorithms and evaluated their performance empirically against baseline methods.

Massachusetts Institute of Technology Cambridge, MA
Summer Program in Undergraduate Research (SPUR) *Jun - Aug 2025*
Advisor: Ankur Moitra

- Co-developed a polynomial-time algorithm to learn the graph of a sparse Gaussian graphical model from a single Glauber trajectory, with a trajectory-length guarantee independent of mixing time (ICML 2026).
- Served as primary internal technical reviewer for the follow-up submitted to ICLR 2027, assessing its improved theoretical guarantees.

Honors and Awards

International Mathematical Olympiad (IMO) Jul 2022
Gold Medal

Academic Service

International Mathematical Olympiad (IMO) 2025
Grader, Problem 6

Teaching and Outreach

Mathink New South Wales, Australia
Main Course Designer and Teacher *May 2023 - Present*

- Design curricula that introduce students to university-level mathematics.
- Teach remotely and develop course materials for advanced mathematics instruction.

Global Teaching Lab Kigali, Rwanda
Instructor *Jan 2024*

- Taught Olympiad mathematics to 40 local teachers and several hundred students.
- Promoted the Rwandan mathematics program at six local high schools to encourage participation in mathematics competitions.

Research Training

FABRIC Somerset, UK
Program on AI and Reasoning (PAIR), Participant *Jul - Aug 2024*

- Studied convolutional neural networks, transformers, and diffusion models.
- Explored mechanistic interpretability, steering vectors, and defenses against prompt injection.

Selected Coursework

Graduate mathematics. Commutative Algebra; Algebraic Topology/Geometry; Number Theory.

Graduate statistics and machine learning. Statistical Inference; Information Theory; Non-asymptotic Statistics; Algorithms for Inference; Deep Learning.

Undergraduate mathematics and computer science. Algebra; Analysis; Topology; Differential Geometry; Computation Structures; Introduction to Programming.

Technical Skills

Python, C/C++, LaTeX, Git, web design.